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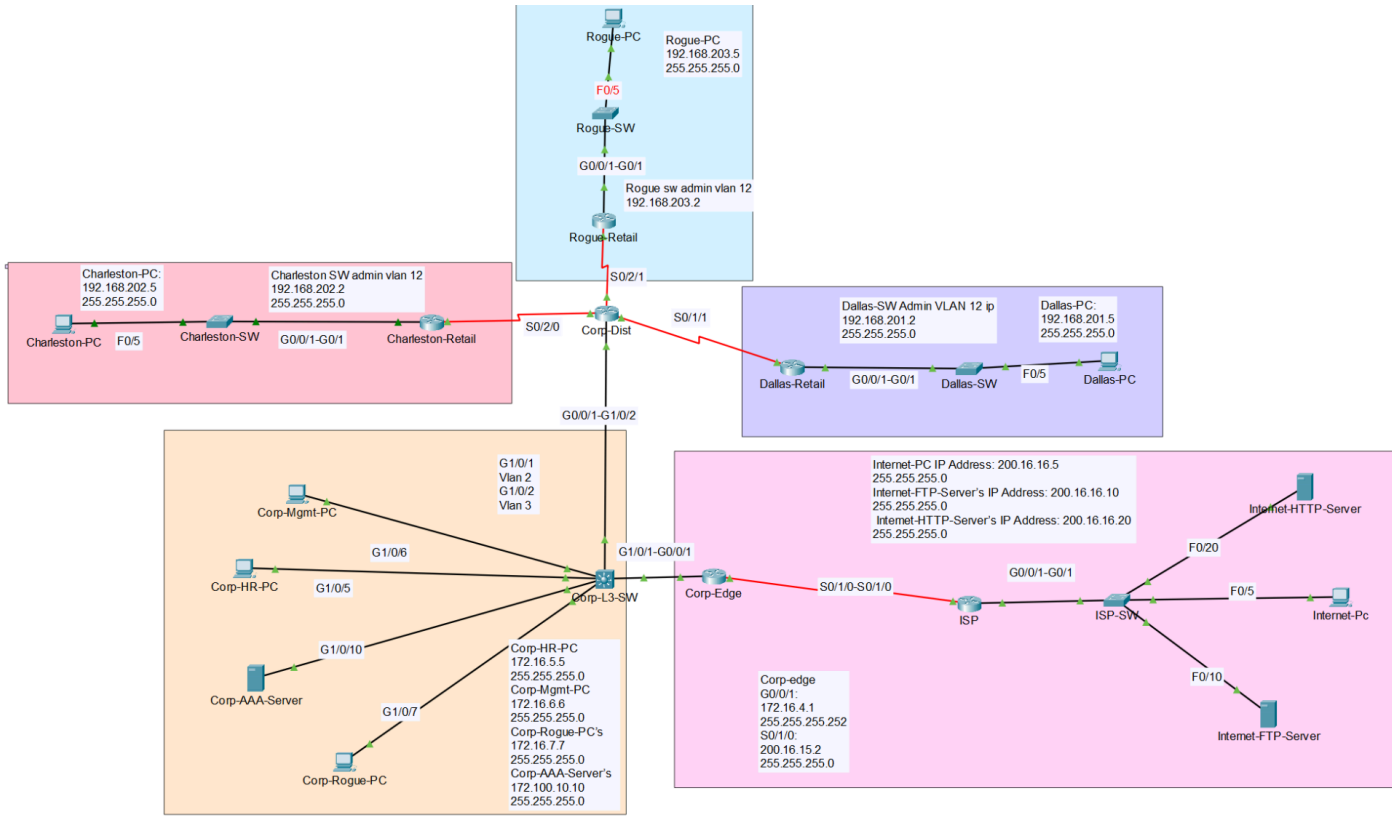
Lab 03
03/29/2024

Description:

This lab focuses on the implementation of secure, server based authentication with AAA, using RADIUS. In this lab we assign usernames and passwords, we begin to limit administrative access to key networking devices, restricting access with ACLs, add AAA services using server based implementations of RADIUS, and protecting and securing these devices and networks.

Topology:



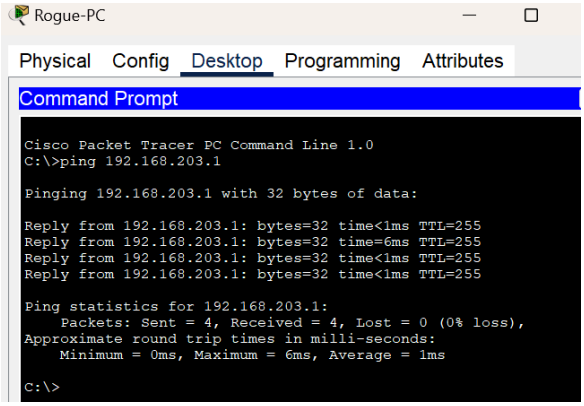
Key Syntax:

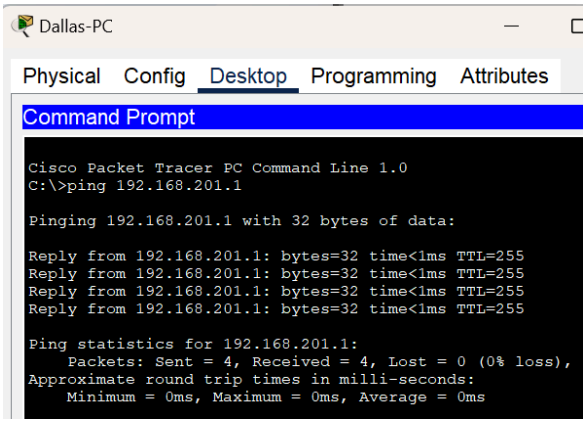
Command	Purpose
hostname *name*	Renames the device in the CLI
Enable Secret *password*	creates an encrypted password
ip address *ip address subnet mask*	sets an interfaces ip address and subnet mask
clock rate 2000000	sets the clock rate for an interface
line console 0	it is used t enter line config mode for the console port
logging synchronous	synchs the log messages with the command input
password *password*	creates a pasword
login	enables checking at login
line vty 0 4	used to enter line config mode for the first five virtual terminal lines
copy runnign-config startup-config	saves the current configurations to the startup config file
router ospf 1	enters the router config mode for OSPF
netword *networkip wildcardmask* area *area*	adds the specified network address to the OSPF area
area *area* authentication message-digest	enables MD5 authentacition
ip ospf messgae-digest-key 1 md5 *key*	sets the OSPF message digest key for the interface to what you set as the key
ip domain-name *name*	creates and names a domain
crypto key generate rsa 2048	generates RSA keys used for SSH
ip ssh version 2	sets SSH version to 2
access-list *number* permit/deny tcp host *host ip* any eq 22	creates an aCL entry that either permits or denies TCP traffic from specified host to any destination port (22)
access-list *number* deny tcp any any eq 2	command creates an ACL entry that denies TCP traffic from any source to any destination on port 22
access-list *number* permit ip any any	reates an ACL entry that permits IP traffic from any source to any destination
access-class *number* in	applies the specified ACL to incoming connections on the line
ip route 0.0.0.0 0.0.0.0 *next hop ip*	sets a default static rute with specified next hop ip address

username *username* password *password*	creates a local user account with the specifies username and password
login local	configs the line to use the local username and password for authenticaion
transport input ssh	restricts the line to only allow SSH connections
access-list 101 permit tcp 172.16.0.0 0.0.255.255 any eq www	this creates an ACL permitting HTTP traffic from the corporate network
access-list 101 deny ip any any	Denies traffic from networks not in the corperate network
ip access-group 101 out/in	this applies the ACL to outbound/inbound interfaces
aaa new model	creates a new aaa model
aaa authentication login default group radius local	configures the login defaults
radius-server host 192.168.1.100 key	configures aaa on the ip address
username aaa password aaa	creates a username and password

Verification:

PCs are able to ping default gateway





Charleston-PC

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.202.1

Pinging 192.168.202.1 with 32 bytes of data:

Reply from 192.168.202.1: bytes=32 time=2ms TTL=255
Reply from 192.168.202.1: bytes=32 time<1ms TTL=255
Reply from 192.168.202.1: bytes=32 time<1ms TTL=255
Reply from 192.168.202.1: bytes=32 time=10ms TTL=255

Ping statistics for 192.168.202.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms
```

Corp-Mgmt-PC

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>
C:\>ping 172.16.6.1

Pinging 172.16.6.1 with 32 bytes of data:

Reply from 172.16.6.1: bytes=32 time<1ms TTL=255
Reply from 172.16.6.1: bytes=32 time<1ms TTL=255
Reply from 172.16.6.1: bytes=32 time<1ms TTL=255
Reply from 172.16.6.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.6.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Corp-Rogue-PC

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.7.1

Pinging 172.16.7.1 with 32 bytes of data:

Reply from 172.16.7.1: bytes=32 time<1ms TTL=255
Reply from 172.16.7.1: bytes=32 time<1ms TTL=255
Reply from 172.16.7.1: bytes=32 time<1ms TTL=255
Reply from 172.16.7.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.7.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Corp-HR-PC

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 172.16.5.1

Pinging 172.16.5.1 with 32 bytes of data:

Reply from 172.16.5.1: bytes=32 time<1ms TTL=255
Reply from 172.16.5.1: bytes=32 time<1ms TTL=255
Reply from 172.16.5.1: bytes=32 time<1ms TTL=255
Reply from 172.16.5.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.5.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Corp edge and isp router ip route table before configuring ustream static route]

Corp-Edge

Physical Config CLI Attributes

IOS Command Line Interface

```
UNAUTHORIZED ACCESS STRICTLY PROHIBITED!!!!$
banner motd

Corp-Edge-Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF
inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA
external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2,
E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia
- IS-IS inter area
       * - candidate default, U - per-user static route, o -
ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.4.0/30 is directly connected,
GigabitEthernet0/0/1
L       172.16.4.1/32 is directly connected,
GigabitEthernet0/0/1
     200.16.15.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.16.15.0/24 is directly connected, Serial0/1/0
L       200.16.15.2/32 is directly connected, Serial0/1/0

Corp-Edge-Router>
```

ISP

Physical Config CLI Attributes

IOS Command Line Interface

```
ISP-Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF
inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA
external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2,
E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia
- IS-IS inter area
       * - candidate default, U - per-user static route, o -
ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     200.16.15.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.16.15.0/24 is directly connected, Serial0/1/0
L       200.16.15.1/32 is directly connected, Serial0/1/0
     200.16.16.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.16.16.0/24 is directly connected,
GigabitEthernet0/0/1
L       200.16.16.1/32 is directly connected,
GigabitEthernet0/0/1

ISP-Router#
```

After configuring upstream static route

Corp-Edge

Physical Config CLI Attributes

IOS Command Line Interface

```
Corp-Edge-Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF
inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E
- EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
IS-IS inter area
       * - candidate default, U - per-user static route, o -
ODR
       P - periodic downloaded static route

Gateway of last resort is 200.16.15.1 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.4.0/30 is directly connected,
GigabitEthernet0/0/1
L       172.16.4.1/32 is directly connected,
GigabitEthernet0/0/1
     200.16.15.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.16.15.0/24 is directly connected, Serial0/1/0
L       200.16.15.2/32 is directly connected, Serial0/1/0
S*   0.0.0.0/0 [1/0] via 200.16.15.1

Corp-Edge-Router#
```

ISP

Physical Config CLI Attributes

IOS Command Line Interface

```
ISP-Router(config)#ip route 0.0.0.0 0.0.0.0 200.16.15.2
ISP-Router(config)#exit
ISP-Router#
%SYS-5-CONFIG_I: Configured from console by console

ISP-Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF
inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA
external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2,
E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia
- IS-IS inter area
       * - candidate default, U - per-user static route, o -
ODR
       P - periodic downloaded static route

Gateway of last resort is 200.16.15.2 to network 0.0.0.0

     200.16.15.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.16.15.0/24 is directly connected, Serial0/1/0
L       200.16.15.1/32 is directly connected, Serial0/1/0
     200.16.16.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.16.16.0/24 is directly connected,
GigabitEthernet0/0/1
L       200.16.16.1/32 is directly connected,
GigabitEthernet0/0/1
S*   0.0.0.0/0 [1/0] via 200.16.15.2
```

Now there are static routes on the routers

On each of the oruters, issue a show ip route command

- Dallas router

```
Dallas-Retail-Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF
inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA
external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2,
E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia
- IS-IS inter area
       * - candidate default, U - per-user static route, o -
ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/30 is subnetted, 1 subnets
O       172.16.4.4/30 [110/65] via 192.168.200.1, 01:55:33,
Serial0/1/1
     192.168.200.0/24 is variably subnetted, 4 subnets, 2
masks
C       192.168.200.0/30 is directly connected, Serial0/1/1
L       192.168.200.2/32 is directly connected, Serial0/1/1
O       192.168.200.4/30 [110/128] via 192.168.200.1,
01:55:16, Serial0/1/1
O       192.168.200.8/30 [110/128] via 192.168.200.1,
01:54:52, Serial0/1/1
     192.168.201.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.201.0/24 is directly connected,
GigabitEthernet0/0/1
L       192.168.201.1/32 is directly connected,
GigabitEthernet0/0/1
O    192.168.202.0/24 [110/129] via 192.168.200.1, 01:55:16,
Serial0/1/1
O    192.168.203.0/24 [110/129] via 192.168.200.1, 01:54:52,
Serial0/1/1
```

```
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia
- IS-IS inter area
       * - candidate default, U - per-user static route, o -
ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.4.4/30 is directly connected,
GigabitEthernet0/0/1
L       172.16.4.6/32 is directly connected,
GigabitEthernet0/0/1
     192.168.200.0/24 is variably subnetted, 6 subnets, 2
masks
C       192.168.200.0/30 is directly connected, Serial0/1/1
L       192.168.200.1/32 is directly connected, Serial0/1/1
C       192.168.200.4/30 is directly connected, Serial0/2/0
L       192.168.200.5/32 is directly connected, Serial0/2/0
C       192.168.200.8/30 is directly connected, Serial0/2/1
L       192.168.200.9/32 is directly connected, Serial0/2/1
O    192.168.201.0/24 [110/65] via 192.168.200.2, 01:56:16,
Serial0/1/1
O    192.168.202.0/24 [110/65] via 192.168.200.6, 01:56:06,
Serial0/2/0
O    192.168.203.0/24 [110/65] via 192.168.200.10, 01:55:44,
Serial0/2/1

Corp-Dist-Router#
```

-Charleston router

```
Gateway of last resort is not set

     172.16.0.0/30 is subnetted, 1 subnets
O       172.16.4.4/30 [110/65] via 192.168.200.5, 01:56:43,
Serial0/2/0
     192.168.200.0/24 is variably subnetted, 4 subnets, 2
masks
O       192.168.200.0/30 [110/128] via 192.168.200.5,
01:56:43, Serial0/2/0
C       192.168.200.4/30 is directly connected, Serial0/2/0
L       192.168.200.6/32 is directly connected, Serial0/2/0
O       192.168.200.8/30 [110/128] via 192.168.200.5,
01:56:14, Serial0/2/0
O    192.168.201.0/24 [110/129] via 192.168.200.5, 01:56:43,
Serial0/2/0
     192.168.202.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.202.0/24 is directly connected,
GigabitEthernet0/0/1
L       192.168.202.1/32 is directly connected,
GigabitEthernet0/0/1
O    192.168.203.0/24 [110/129] via 192.168.200.5, 01:56:14,
Serial0/2/0

Charleston-Retail-Router#
```

```
Gateway of last resort is not set

     172.16.0.0/30 is subnetted, 1 subnets
O       172.16.4.4/30 [110/65] via 192.168.200.9, 01:56:45,
Serial0/2/1
     192.168.200.0/24 is variably subnetted, 4 subnets, 2
masks
O       192.168.200.0/30 [110/128] via 192.168.200.9,
01:56:45, Serial0/2/1
O       192.168.200.4/30 [110/128] via 192.168.200.9,
01:56:45, Serial0/2/1
C       192.168.200.8/30 is directly connected, Serial0/2/1
L       192.168.200.10/32 is directly connected, Serial0/2/1
O    192.168.201.0/24 [110/129] via 192.168.200.9, 01:56:45,
Serial0/2/1
O    192.168.202.0/24 [110/129] via 192.168.200.9, 01:56:45,
Serial0/2/1
     192.168.203.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.203.0/24 is directly connected,
GigabitEthernet0/0/1
L       192.168.203.1/32 is directly connected,
GigabitEthernet0/0/1

Rouge-Retail-Router#
```

-corp edge router and ISP router's ip route

IOS Command Line Interface

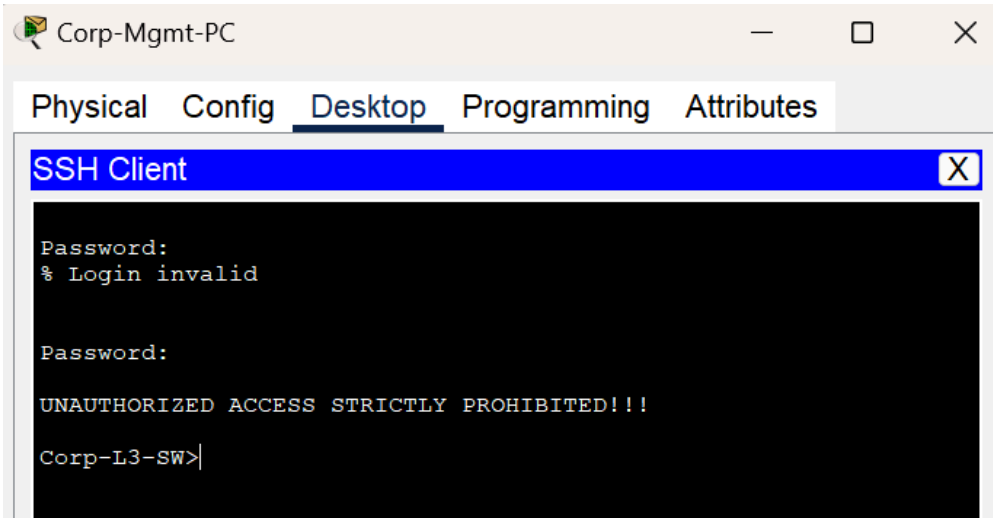
```
Corp-Edge-Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF
inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E
- EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
IS-IS inter area
       * - candidate default, U - per-user static route, o -
ODR
       P - periodic downloaded static route

Gateway of last resort is 200.16.15.1 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.4.0/30 is directly connected,
GigabitEthernet0/0/1
L       172.16.4.1/32 is directly connected,
GigabitEthernet0/0/1
     200.16.15.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.16.15.0/24 is directly connected, Serial0/1/0
L       200.16.15.2/32 is directly connected, Serial0/1/0
S*   0.0.0.0/0 [1/0] via 200.16.15.1

Corp-Edge-Router#
```

```
ISP-Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF
inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA
external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2,
E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia
- IS-IS inter area
       * - candidate default, U - per-user static route, o -
ODR
       P - periodic downloaded static route

Gateway of last resort is 200.16.15.2 to network 0.0.0.0

     200.16.15.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.16.15.0/24 is directly connected, Serial0/1/0
L       200.16.15.1/32 is directly connected, Serial0/1/0
     200.16.16.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.16.16.0/24 is directly connected,
GigabitEthernet0/0/1
L       200.16.16.1/32 is directly connected,
GigabitEthernet0/0/1
S*   0.0.0.0/0 [1/0] via 200.16.15.2

ISP-Router#
```

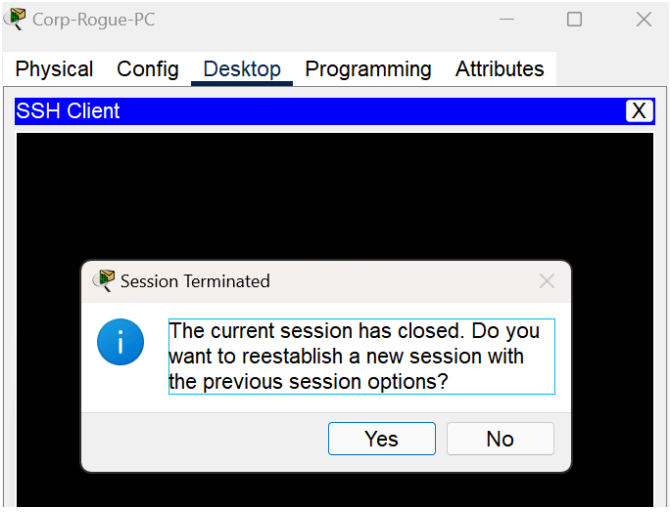
```
Charleston-Retail-Router>enable
Password:
Charleston-Retail-Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Charleston-Retail-Router(config)#username cisco secret cisco
Charleston-Retail-Router(config)#username julianna.russo
secret lab03Password
```

Copy Paste

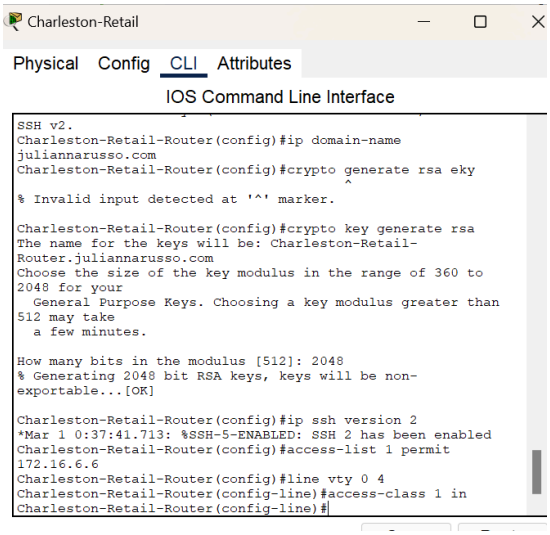
I am able to SSH into switches and routers through Corp-mgmt-pc, but only with the correct username and password



When I try to ssh into a switch or router from any device other than Corp-mgmt-PC i am unable to and get this error message



- This is the configurations for SSH on the routers.



- Configuring AAA

Dallas-Retail

Physical Config CLI Attributes

IOS Command Line Interface

```
Dallas-Retail-Router>enable
Password:
Dallas-Retail-Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dallas-Retail-Router(config)#
Dallas-Retail-Router(config)#radius server 172.100.10.10
Dallas-Retail-Router(config-radius-server)#key 12
 WARNING: Command has been added to the configuration using a
type 0 password. However, type 0 passwords will soon be
deprecated. Migrate to a supported password type
*Mar 28 20:30:46.531: %AAAA-4-CLI_DEPRECATED: WARNING: Command
has been added to the configuration using a type 0 password.
However, type 0 passwords will soon be deprecated. Migrate to a
supported password type
Dallas-Retail-Router(config-radius-server)#
Dallas-Retail-Router(config-radius-server)#username aaa password
aaa
Dallas-Retail-Router(config)#
Dallas-Retail-Router(config)#username aaa-julianna password
juliannarussolab3
Dallas-Retail-Router(config)#
Dallas-Retail-Router(config)#aaa new-model
Dallas-Retail-Router(config)#
Dallas-Retail-Router(config)#aaa authentication login default
group radius local
Dallas-Retail-Router(config)#exit
Dallas-Retail-Router#
%SYS-5-CONFIG_I: Configured from console by console

Dallas-Retail-Router#exit
```

ISP

Physical Config CLI Attributes

IOS Command Line Interface

```
UNAUTHORIZED ACCESS STRICTLY PROHIBITED!!!!!!

ISP-Router>enable
Password:
ISP-Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ISP-Router(config)#
ISP-Router(config)#radius server 172.100.10.10
ISP-Router(config-radius-server)#key 12
 WARNING: Command has been added to the configuration using a
type 0 password. However, type 0 passwords will soon be
deprecated. Migrate to a supported password type
*Mar 28 20:31:30.140: %AAAA-4-CLI_DEPRECATED: WARNING:
Command has been added to the configuration using a type 0
password. However, type 0 passwords will soon be deprecated.
Migrate to a supported password type
ISP-Router(config-radius-server)#
ISP-Router(config-radius-server)#username aaa password aaa
ISP-Router(config)#
ISP-Router(config)#username aaa-julianna password
juliannarussolab3
ISP-Router(config)#
ISP-Router(config)#aaa new-model
ISP-Router(config)#
ISP-Router(config)#aaa authentication login default group
radius local
ISP-Router(config)#exit
ISP-Router#
%SYS-5-CONFIG_I: Configured from console by console

ISP-Router#exit
```

The commands to only allow http

Corp-Edge

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0,
changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0,
changed state to up

Corp-Edge-Router>config t
                 ^
% Invalid input detected at '^' marker.

Corp-Edge-Router>enable
Password:
Corp-Edge-Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Corp-Edge-Router(config)#access-list 101 permit tcp
172.16.0.0 0.0.255.255 any eq www
Corp-Edge-Router(config)#access-list 101 deny ip any any
Corp-Edge-Router(config)#
Corp-Edge-Router(config)#interface GigabitEthernet0/0
%Invalid interface type and number
Corp-Edge-Router(config)# ip access-group 101 out
                         ^
% Invalid input detected at '^' marker.

Corp-Edge-Router(config)#int s0/1/0
Corp-Edge-Router(config-if)# ip access-group 101 out
Corp-Edge-Router(config-if)#
```

These are some of the final cofiguratins on the devices in the network

ISP-SW

Physical Config CLI Attributes

IOS Command Line Interface

```
!
ip default-gateway 200.16.16.1
!
banner motd ^CUNAUTHORIZED ACCESS STRICTLY PROHIBITED !!!^C
!
!
!
access-list 1 permit host 172.16.6.6
line con 0
 logging synchronous
 login
!
line vty 0 4
 access-class 1 in
 logging synchronous
 login local
 transport input ssh
line vty 5 15
 login
!
!
!
!
end
```

Rogue-Retail

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!

UNAUTHORIZED ACCESS STRICLY PROHIBITED !!!

User Access Verification

Username: julianna.russo
Password:

Rouge-Retail-Router>enable
Password:
Rouge-Retail-Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Rouge-Retail-Router(config)#show access-list
                                           ^
% Invalid input detected at '^' marker.

Rouge-Retail-Router(config)#do sh access-list
Standard IP access list 1
    10 permit host 172.16.6.6
Extended IP access list 101
    10 permit tcp host 172.16.6.6 any eq 22

Rouge-Retail-Router(config)#
```

Corp-Edge

Physical Config CLI Attributes

IOS Command Line Interface

```
UNAUTHORIZED ACCESS STRICTLY PROHIBITED!!!!$
banner motd

User Access Verification

Username: julianna.russo
Password:

Corp-Edge-Router>enable
Password:
Corp-Edge-Router#sh acl
                   ^
% Invalid input detected at '^' marker.

Corp-Edge-Router#sh access-list
Standard IP access list 1
    10 permit host 172.16.6.6 (4 match(es))
Extended IP access list 101
    10 permit tcp host 172.16.6.6 any eq 22
    20 permit tcp 172.16.0.0 0.0.255.255 any eq www
    30 deny ip any any

Corp-Edge-Router#
```

Corp-L3-SW

Physical Config CLI Attributes

```
no service timestamps debug datetime msec
service password-encryption
!
hostname Corp-L3-SW
!
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
aaa new-model
!
aaa authentication login default group radius local
!
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
!
!
!
username aaa password 7 08204D4F
username aaa-julianna password 7 082B594200180B19131919173924282931 66
username cisco password 7 0822455D0A16
username julianna.russo password 7 082B594200180B19131919173924082931 66
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name juliannarusso.com
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet1/0/1
 switchport access vlan 2
 switchport mode access
 switchport nonegotiate
!
```

```
!
interface GigabitEthernet1/0/1
 switchport access vlan 2
 switchport mode access
 switchport nonegotiate
!
interface GigabitEthernet1/0/2
 switchport access vlan 3
 switchport mode access
 switchport nonegotiate
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
 switchport access vlan 5
 switchport mode access
 switchport nonegotiate
!
interface GigabitEthernet1/0/6
 switchport access vlan 6
 switchport mode access
 switchport nonegotiate
!
interface GigabitEthernet1/0/7
 switchport access vlan 7
 switchport mode access
 switchport nonegotiate
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
 switchport access vlan 10
 switchport mode access
 switchport nonegotiate
!
```

```
 shutdown
!
interface Vlan2
 mac-address 0010.1182.4501
 ip address 172.16.4.2 255.255.255.252
!
interface Vlan3
 mac-address 0010.1182.4502
 ip address 172.16.4.5 255.255.255.252
!
interface Vlan5
 mac-address 0010.1182.4503
 ip address 172.16.5.1 255.255.255.0
 ip ospf message-digest-key 1 md5 cisco
!
interface Vlan6
 mac-address 0010.1182.4504
 ip address 172.16.6.1 255.255.255.0
 ip ospf message-digest-key 1 md5 cisco
!
interface Vlan7
 mac-address 0010.1182.4505
 ip address 172.16.7.1 255.255.255.0
 ip ospf message-digest-key 1 md5 cisco
!
interface Vlan10
 mac-address 0010.1182.4506
 ip address 172.16.10.1 255.255.255.0
 ip ospf message-digest-key 1 md5 cisco
!
interface Vlan20
 mac-address 0010.1182.4507
 no ip address
!
router ospf 1
 router-id 172.16.4.1
 log-adjacency-changes
 area 0 authentication message-digest
 network 172.16.5.0 0.0.0.255 area 0
 network 172.16.6.0 0.0.0.255 area 0
 network 172.16.7.0 0.0.0.255 area 0
 network 172.16.10.0 0.0.0.255 area 0
 network 172.16.4.0 0.0.0.3 area 0
 default-information originate
!
ip classless
!
ip flow-export version 9
!
!
access-list 1 permit host 172.16.6.6
!
banner motd ^CUNAUTHORIZED ACCESS STRICTLY PROHIBITED!!!^C
!
!
!
!
line con 0
 logging synchronous
```

```
.
access-list 1 permit host 172.16.6.6
!
banner motd ^CUNAUTHORIZED ACCESS STRICTLY PROHIBITED!!!^C
!
!
!
!
line con 0
 logging synchronous
!
line aux 0
!
line vty 0 4
 access-class 1 in
 logging synchronous
 transport input ssh
!
!
!
!
end
```

Rogue-SW

Physical Config CLI Attributes

IOS Command Line Interface

```
 shutdown
!
interface FastEthernet0/24
 shutdown
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan12
 ip address 192.168.203.2 255.255.255.0
!
interface Vlan13
 no ip address
 shutdown
!
ip default-gateway 192.168.203.1
!
banner motd ^CUNAUTHORIZED ACCESS STRICTLY PROHIBITED!!!^C
!
!
!
access-list 1 permit host 172.16.6.6
line con 0
 logging synchronous
 login
!
line vty 0 4
 access-class 1 in
 logging synchronous
 login local
 transport input ssh
line vty 5 15
 login
!
```

Rogue-SW

Physical Config CLI Attributes

IOS Command Line Interface

```
hostname Rogue-SW
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
ip ssh version 2
ip domain-name juliannarusso.com
!
username cisco privilege 1 password 7 0822455D0A16
username julianna.russo privilege 1 password 7
082B594200180B1913191917392408293166
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 shutdown
!
interface FastEthernet0/2
 shutdown
!
interface FastEthernet0/3
 shutdown
!
interface FastEthernet0/4
 shutdown
!
interface FastEthernet0/5
!
interface FastEthernet0/6
 shutdown
!
```

Corp-Edge

Physical Config CLI Attributes

```
UNAUTHORIZED ACCESS STRICTLY PROHIBITED!!!!$
banner motd

User Access Verification

Username: julianna.russo
Password:
Corp-Edge-Router>enable
Password:
Corp-Edge-Router#sh run
Building configuration...

Current configuration : 1731 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Corp-Edge-Router
!
!
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
aaa new-model
!
aaa authentication login default group radius local
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username aaa password 7 08204D4F
username aaa-julianna password 7 082B594200180B19131919173924082931 66
username aaa-julianna.russo password 7 082B594200180B1913191917392408293166
username cisco password 7 0822455D0A16
username julianna.russo password 7 082B594200180B1913191917392408293166
!
!
```

Corp-Edge

Physical Config CLI Attributes

```
ip ssh version 2
ip domain-name juliannarusso.com
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/0/1
 ip address 172.16.4.1 255.255.255.252
 duplex auto
 speed auto
!
interface Serial0/1/0
 ip address 200.16.15.2 255.255.255.0
 ip access-group 101 out
!
interface Serial0/1/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 172.16.4.0 0.0.0.3 area 0
 network 200.16.15.0 0.0.0.255 area 0
!
ip classless
ip route 0.0.0.0 0.0.0.0 200.16.15.1
!
ip flow-export version 9
!
!
access-list 1 permit host 172.16.6.6
access-list 101 permit tcp host 172.16.6.6 any eq 22
access-list 101 permit tcp 172.16.0.0 0.0.255.255 any eq www
access-list 101 deny ip any any
!
banner motd ^CUNAUTHORIZED ACCESS STRICTLY PROHIBITED!!!!$
banner motd ^C
!
radius server 172.100.10.10
 key 12
!
!
!
line con 0
```

```
line con 0
 logging synchronous
!
line aux 0
!
line vty 0 4
 access-class 101 in
 logging synchronous
 transport input ssh
!
!
!
end


Corp-Edge-Router#
```

Conclusion:

This lab focuses on different types of protections to secure networks, besides just usernames and passwords. AAA, SSH, and radius were focused on, and the use of access control lists were also implemented.